

AI Project Logbook

TrustCheck: Fake Review Detection & Trust Score System

PROJECT NAME:

SCHOOL NAME: Cambridge Court World School

YEAR/CLASS 2025-26 / Class 12 A

TEACHER NAME: Ms. Priyanka Sharma

TEACHER EMAIL: comp5@cambridgecourtworldschool.org

TEAM MEMBER NAMES AND GRADES:

1. Deshna (Class 12 A) ? Group Leader, Data Expert, Prototype Builder
2. Aryan (Class 12 A) ? Tester, Designer
3. Bhavya (Class 12 A) ? Information Researcher
4. Darshil (Class 12 A) ? Video Producer
5. Charvi (Class 12 A) ? Marketing / Communication Leader

1. Prepare for the project

This document is your **Project Logbook**, and it will be where you record your ideas, thoughts and answers as you work to solve a local problem using AI.

Make a copy of the document in your shared drive and work through it digitally with your team. You can also print a copy of the document and submit a scanned copy once you have completed the Project Logbook. Feel free to add pages and any other supporting material to this document.

Refer to the **AI Project Guide** for more details about what to do at each step of your project.

2. Team Roles

2.1 Who is in your team and what are their roles?

Role	Role description	Team Member Name
Project leader	• Schedules the task among the team member, • Ensures the task is completed on time, • Resolves doubts (if any), and one source of contact	Deshna
Data expert	• Decides upon the data required, type of data for training the model, • collects the data, • ensures the data is in a format that the team can work with • ensures the data is ethically sourced and unfair bias is eliminated. • Works with prototype builder to train the AI model.	Deshna
Information researcher	• Collects questions from the team which needs to be answered, • look for the answers for those questions and then prepare a report which will be passed to the project leader	Bhavya

Designer	- Will create the design and the flow of how to go about making the solution for the problem statement. - Plans the user interface for the prototype	Aryan
Prototype builder/coder	Works to build the model, train it followed by testing the efficiency and accuracy of the model	Deshna
Tester	- Works with user to test the prototype. - Gets feedback from user and user signoff when the prototype has met user requirements . - Create an action plan on what needs to be fixed and prioritizes requests for future improvements.	Aryan
Marketing/ Communication Leader	- Collats the team project logbook submission and creates the content for the video pitch. - Selects spokespeople within the team for various matters related to the project.	Charvi
Video Producer	Films the activities of the team and edits these into a presentation for submission.	Darshil

2.2 Project plan

The following table is a guide for your project plan. You may use this or create your own version using a spreadsheet which you can paste into this section. You can expand the 'Notes' section to add reminders, things that you need to follow up on, problems that need to be fixed urgently, etc.

Phase	Task	Planned start date	Planned end date	Planned duration (hours, minutes)	Actual start date	Actual end date	Actual duration (hours, minutes)	Who is responsible	Notes/Remarks
Preparing for the project	Coursework, readings	04/08/25	04/08/25	1 Hour	04/08/25	04/08/25	1 Hour	Team Members	Each member is responsible
Forming the team	Getting to know the people in the team	05/08/25	05/08/25	1 Hour	05/08/25	05/08/25	1 Hour	Team Members	Defined roles & guidelines

Defining the problem	Background reading	07/08/25	07/08/25	2 Hours	07/08/25	07/08/25	2 Hours	Team Members	Collaborative work
	Research issues in our community	08/08/25	08/08/25	2 Hours	08/08/25	08/08/25	2 Hours	Team Members	Surveyed students
	Team meeting to discuss issues and select an issue for the project	09/08/25	09/08/25	2 Hours	09/08/25	09/08/25	2 Hours	Team Members	Selected fake reviews
	Complete section 3 of the Project Logbook	10/08/25	10/08/25	15 Mins	10/08/25	10/08/25	15 Mins	Team Members	Problem defined
	Rate yourselves	3/3							
Understanding the users	Identify users	12/08/25	12/08/25	30 Mins	12/08/25	12/08/25	30 Mins	Team Members	Online interactions with the end user
	Meeting with users to observe them	13/08/25	13/08/25	30 Mins	13/08/25	13/08/25	30 Mins	Team Members	Amazon shoppers
	Interview with user (1)	14/08/25	14/08/25	15 Mins	14/08/25	14/08/25	15 Mins	Team Members	Class 12 student
	Interview with user (2), etc....	14/08/25	14/08/25	15 Mins	14/08/25	14/08/25	15 mins	Team Members	School teacher
	Complete section 4 of the Project Logbook	15/08/25	15/08/25	30 Mins	15/08/25	15/08/25	30 Mins	Team Members	Collaborative work
	Rate yourselves	3/3							

Brainstorming	Team meeting to generate ideas for a solution	17/08/25	17/08/25	1.5 Hours	17/08/25	17/08/25	1.5 Hours	Team Members	ML vs Rules
	Complete section 5 of the Project Logbook	18/08/25	18/08/25	30 Mins	18/08/25	18/08/25	30 Mins	Team Members	Collaborative work
	Rate yourselves	3/3							
Designing your solution	Team meeting to design the solution	20/08/25	20/08/25	3.5 Hours	20/08/25	20/08/25	3.5 Hours	Team Members	UI & Trust math
	Complete section 6 of the logbook	21/08/25	21/08/25	30 mins	21/08/25	21/08/25	30 Mins	Team Members	Collaborative work
	Rate yourselves	3/3							
Collecting and preparing data	Team meeting to discuss data requirements	23/08/25	23/08/25	30 Mins	23/08/25	23/08/25	30 Mins	Team Members	Data requirements
Collecting and preparing data Prototyping	Data collection	24/08/25	24/08/25	1 Hour	24/08/25	24/08/25	1 Hour	Deshna	Kaggle 40k dataset
	Data preparation and labelling	25/08/25	25/08/25	2 Hours	25/08/25	25/08/25	2 Hours	Team Members	TF-IDF 80/20 split
	Complete Section 6 of the Project Logbook	26/08/25	26/08/25	30 Mins	26/08/25	26/08/25	30 Mins	Team Members	Collaborative work
	Team meeting to plan prototyping phase	28/08/25	28/08/25	4 Hours	28/08/25	28/08/25	4 Hours	Team Members	Code testing

Prototyping Testing	Train your model with input dataset	30/08/25	30/08/25	1 Hour	30/08/25	30/08/25	1 Hour	Team Members	Designing
	Test your model and keep training with more data until you think your model is accurate	01/09/25	01/09/25	30 Mins	01/09/25	01/09/25	30 Mins	Team Members	Code testing
	Write a program to initiate actions based on the result of your model	03/09/25	03/09/25	45 Mins	03/09/25	03/09/25	45 Mins	Team Members	Trust Score API
	Complete section 8 of the Project Logbook	04/09/25	04/09/25	15 Mins	04/09/25	04/09/25	15 Mins	Team Members	Collaborative work
	Rate yourselves	3/3							
	Team meeting to discuss testing plan	06/09/25	06/09/25	10 Mins	06/09/25	06/09/25	10 Mins	Team Members	Collaborative work
Testing Creating the video	Invite users to test your prototype	07/09/25	07/09/25	15 Mins	07/09/25	07/09/25	15 mins	Team Members	12 test users
	Conduct testing with users	08/09/25	08/09/25	30 Mins	08/09/25	08/09/25	30 Mins	Team Members	Live link test
	Complete section 9 of the Project Logbook	09/09/25	09/09/25	30 mins	09/09/25	09/09/25	10 Mins	Team Members	Collaborative work
	Rate yourselves	3/3							

	Team meeting to discuss video creation	11/09/25	11/09/25	10 Mins	11/09/25	11/09/25	10 Mins	Team Members	Collaborative work
	Write your script	12/09/25	12/09/25	20 Mins	12/09/25	12/09/25	20 Mins	Team Members	Pitch script
	Film your video	13/09/25	13/09/25	30 Mins	13/09/25	13/09/25	30 Mins	Team Members	Recorded demo
	Edit your video	14/09/25	14/09/25	30 Mins	14/09/25	14/09/25	30 Mins	Team Members	Voiceover added
Completing the logbook	Reflect on the project with your team	16/09/25	16/09/25	10 Mins	16/09/25	16/09/25	10 Mins	Team Members	
	Complete sections 10 and 11 of the Project Logbook	17/09/25	17/09/25	20 Mins	17/09/25	17/09/25	20 Mins	Team Members	Collaborative work
	Review your Project logbook and video	18/09/25	18/09/25	10 Mins	18/09/25	18/09/25	10 Mins	Team Members	
Submission	Submit your entries on the IBM	20/09/25	20/09/25	10 Mins	20/09/25	20/09/25	10 Mins	Team Members	Individual on google classroom

2.3 Communications plan

Will you meet face-to-face, online or a mixture of each to communicate?

How often will you come together to share your progress?

Who will set up online documents and ensure that everyone is contributing?

What tools will you use for communication?

? Meeting Format: A hybrid mixture of daily face-to-face check-ins in the school computer lab and virtual evening sessions.

? Meeting Frequency: 15-minute daily standups after classes (Mon-Fri) and 45-minute sprint reviews every Saturday.

? Online Document Management: Deshna (Project Leader) set up shared Google Drive & GitHub; Charvi monitored task progress.

? Communication Tools: Google Meet (virtual sprints), WhatsApp Group (daily alerts), Google Docs (logbook), GitHub (code).

2.4 Team meeting minutes (create one for each meeting held)

Date of meeting: 09/08/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None (Full team present)

Purpose of meeting: Brainstorm community issues & select the core AI project problem.

Items discussed:

1. Discussed community feedback on e-commerce scams and misleading product star ratings.
2. Confirmed fake 5-star reviews on Amazon/Flipkart trick students and parents into buying defective items.
3. Decided to build TrustCheck using Natural Language Processing and Machine Learning to detect deceptive reviews.

Things to do (what, by whom, by when):

1. Literature review on review spam detection algorithms ? Bhavya ? by 11/08/2025.
2. Draft student and parent survey questionnaire ? Charvi ? by 12/08/2025.
3. Search and evaluate open labeled datasets on Kaggle ? Deshna ? by 12/08/2025.

Date of meeting: 15/08/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None

Purpose of meeting:

To finalize the problem statement and system objectives.

Items discussed:

1. Reviewed survey results from 12 school community shoppers; 85% fell victim to inflated ratings.
2. Star ratings are broken because platforms compute a plain arithmetic average including bot reviews.
3. Defined primary goals: Genuine vs Fake ML classifier, real Adjusted Rating, and a 0-100 Trust Score.

Things to do (what, by whom, by when):

1. Complete sections 3 and 4 of Project Logbook ? Deshna & Charvi ? by 16/08/2025.
2. Design initial UI wireframes and user interaction flow ? Aryan ? by 18/08/2025.
3. Research TF-IDF vectorization and classification models ? Bhavya ? by 19/08/2025.

Date of meeting: 20/08/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None

Finalize solution design, UI layout, and mathematical Trust Score formula.

Purpose of meeting:

1. Finalized calm, nature-toned SaaS UI: side-by-side star comparison and circular Trust Score gauge.
2. Formulated Trust Score equation: $\text{Trust Score} = 100 - (\text{Fake \%} \times \text{Confidence} / 100)$.

Items discussed:

3. Defined 3 input channels: direct review paste, CSV batch upload, and live storefront URLs.

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1. Develop HTML/Tailwind CSS frontend components ? Aryan ? by 23/08/2025.

Things to do (what, by whom, by when):

2. Download and verify Kaggle 40,432 balanced fake reviews dataset ? Deshna ? by 24/08/2025.
3. Draft outline for video pitch and demonstration script ? Charvi & Darshil ? by 25/08/2025.

Date of meeting: 25/08/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None

Purpose of meeting: Data preprocessing, feature engineering, and model selection.

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Items discussed:

1. Preprocessed 40,432 reviews: text cleaning, punctuation removal, and stop-word filtering.
2. Extracted top 5,000 unigram/bigram TF-IDF features to capture deception markers (e.g. 'must buy').
3. Outlined benchmark experiment: Logistic Regression vs Multinomial Naive Bayes.

Things to do (what, by whom, by when):

1. Write train_model.py script and compute accuracy/F1-scores ? Deshna ? by 29/08/2025.
2. Audit dataset balance and confirm safety/privacy compliance ? Bhavya ? by 28/08/2025.
3. Prepare test cases and real-world product review samples ? Aryan ? by 30/08/2025.

Date of meeting: 03/09/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None

Purpose of meeting: Prototype integration (connecting Flask ML backend with Web UI).

Items discussed:

1. Model training successful: Logistic Regression reached 89.4% accuracy (outperformed Naive Bayes at 85.2%).
2. Connected Flask REST API endpoints (/analyze and /analyze-single) with sub-50ms inference.
3. Implemented 4-vector forensics engine (Language, Template, Sentiment, and Behavior risk factors).

Things to do (what, by whom, by when):

1. Deploy full-stack application live on Vercel platform ? Deshna & Aryan ? by 05/09/2025.
2. Conduct user testing sessions with 12 students and teachers ? Aryan & Charvi ? by 07/09/2025.
3. Film behind-the-scenes coding and testing footage for video ? Darshil ? by 06/09/2025.

Date of meeting: 11/09/2025

Who attended: Deshna, Aryan, Bhavya, Darshil, Charvi

Who wasn't able to attend: None

Purpose of meeting: User testing review, UI refinement, and video production plan.

Items discussed:

- 1.
1. Reviewed user feedback: users praised the Dual-Rating feature but requested simpler mobile navigation.
2. Fixed duplicate navigation buttons on dashboard and added graceful CAPTCHA fallback notice.
3. Storyboarded 3-minute video pitch: Problem, Hook -> TrustCheck Live Demo -> AI Pipeline -> Impact.

Things to do (what, by whom, by when):

- 1.
1. Record live screen demo and member voiceovers ? Darshil & Team ? by 13/09/2025.
2. Edit final video submission with transitions and subtitles ? Darshil ? by 15/09/2025.
3. Review entire logbook against rubrics and submit ? All Team Members ? by 17/09/2025.

3. Problem Definition

3.1 List important local issues faced by your school or community

1. Misleading E-Commerce Reviews: Online shoppers receive low-quality items because sellers purchase fake 5-star praise.
2. Rating Inflation: Shopping platforms average all reviews blindly, distorting poor products into 4.5+ star items.
3. School Feedback Spam: Anonymous school event and cafeteria forms get flooded with non-genuine repeat text.
4. Small Business Review Sabotage: Honest local vendors face 1-star competitor spam while deceptive stores thrive.

3.2 Which issues matter to you and why?

Deceptive Online Product Reviews matter most because over 93% of our school community relies on customer star ratings before buying. Fake reviews deceive buyers into wasting money on defective electronics and study gear.

3.3 Which issue will you focus on?

We focus on detecting deceptive and computer-generated fake customer reviews on e-commerce platforms (Amazon/Flipkart) and calculating an honest, adjusted Trust Score (0-100) alongside the true genuine star rating.

3.4 Write your team's problem statement in the format below.

How might we empower online shoppers in our community to automatically detect fake product reviews so that they can uncover true product quality and make confident, money-saving purchases based on genuine customer feedback?

Rate yourself

3

Problem Definition

1 point - A local problem is described

2 points - A local problem which has not been fully solved before is described.

3 points - A local problem which has not been fully solved before is explained in detail with upporting research.

4. The Users

4.1 Who are the users and how are they affected by the problem?

Everyday online shoppers, including high school students purchasing gadgets and study tools, parents buying household goods, and teachers ordering classroom supplies. They suffer financial loss, wasted time, and broken trust.

4.2 What have you actually observed about the users and how the problem affects them?

Users instinctively filter products by '4+ stars', scroll past dozens of generic 5-star praise reviews, struggle to spot bot templates manually, and often abandon carts or regret purchases when products turn out defective.

4.3 Record your interview questions here as well as responses from users.

Q: 'Do you check reviews before buying?' User 1: 'Always. But I bought a 4.7? mouse that broke in 2 days? all reviews were fake!'

Q: 'Can you easily spot fake reviews?' User 2: 'It takes 20 minutes to read through 50 reviews and I still get fooled by paid bots.'

4.4 Empathy Map

Map what the users say, think, do and feel about the problem in this table

What our users are saying ? 'The item had 4.8 stars but broke immediately!' ? 'I don't know which reviews to trust anymore.' ? 'Reading 50 reviews is exhausting.'	What our users thinking ? 'Are these reviews written by paid bots?' ? 'Sellers are gaming the system to take my money.' ? 'I wish an AI could verify these reviews for me.'
What our users are doing ? Scrolling past top reviews hunting for 1-star complaints. ? Second-guessing ratings and hesitating to purchase. ? Blindly trusting 5-star ratings and regretting purchases.	How our users feel ? Frustrated by wasted money and return hassles. ? Skeptical and distrustful of e-commerce ratings. ? Overwhelmed by information overload.

4.5 What are the usual steps that users currently take related to the problem and where are the difficulties?

Current steps: Search product -> Check stars -> Skim reviews -> Search external forums -> Blind purchase. Difficulties: Cognitive bias, human inability to spot linguistic deception patterns across 100+ reviews, and heavy time consumption

4.6 Write your team's problem statement in the format below.

Our users need an automated, transparent review verification tool because dishonest sellers flood products with fake 5-star reviews, artificially inflating ratings and leading to financial loss and wasted time.

Rate yourself

3

The Users

1 point - The user group is described but it is unclear how they are affected by the problem.

2 points - Understanding of the user group is evidenced by completion of most of the steps in this section.

3 points - Understanding of the user group is evidenced by completion of most of the steps in this section and thorough investigation

5. Brainstorming

5.1 Ideas

How might you use the power of AI/machine learning to solve the users' problem by increasing their knowledge or improving their skills?

AI Idea #1	Rule-Based Keyword Blacklist Filter: Flags repetitive spam words ('must buy', 'guaranteed'). Easy to build, but easily bypassed by modern bot text and misses nuanced context.
AI Idea #2	Supervised ML Classifier (TrustCheck): TF-IDF + Logistic Regression on 40k+ reviews to detect subtle lexical deception, compute 0-100 Trust Score, and recalculate genuine star rating.
AI Idea #3	Computer Vision Review Authenticator: Verifies customer photos against web images to catch stolen stock photos. Complex, computationally heavy, and ignores text-only reviews.
AI Idea #4	Browser Extension Overlay: Injects Trust Score badges directly onto Amazon and Flipkart product pages. High user convenience, ideal as Phase 2 after building the core ML web application.

5.2 Priority Grid

Evaluate your five AI ideas based on value to users and ease of creation and implementation.

Value to Users	Ease of creation	AI Idea
High	Easy	AI Idea #2: Supervised ML Classifier (TrustCheck)
High	Hard	AI Idea #4: Browser Extension Overlay with Scraping
Low	Easy	AI Idea #1: Rule-Based Keyword Blacklist Filter
Low	Hard	AI Idea #3: Computer Vision Image Authenticator

5.3 Based on the priority grid, which AI solution is the best fit for your users and for your team to create and implement?

Briefly summarize the idea for your solution in a few sentences and be sure to identify the tool that you will use.

AI Idea #2 (TrustCheck) is the best fit. Built with Python, Scikit-Learn (TF-IDF + Logistic Regression), Flask API, and Next.js, achieving 89.4% accuracy to deliver instant Trust Scores and real adjusted ratings.

Rate yourself

3

Brainstorming

1 point – A brainstorming session was conducted. A solution was selected.

2 points - A brainstorming session was conducted using creative and critical thinking. A solution was selected with supporting arguments in this section

3 points - A brainstorming session was conducted using creative and critical thinking. A compelling

6. Design

6.1 What are the steps that users will now do using your AI solution to address the problem?

1. Access Web App: User opens the live TrustCheck application on their browser (trust-check-rouge.vercel.app).
2. Select Input Mode: User enters an Amazon/Flipkart URL, pastes review text, or uploads a batch CSV file.
3. One-Click AI Analysis: System vectorizes text via TF-IDF and runs Logistic Regression prediction in <50ms.
4. Inspect Dual-Rating: User views the Advertised Rating (e.g. 4.6?) right next to the Adjusted Genuine Rating (e.g. 3.8?).
5. Evaluate Trust Score: User checks the 0?100 circular gauge (80-100: Reliable; 50-79: Caution; <50: High Risk).
6. Review 4-Vector Forensics: User examines flagged risk cues (Language, Template, Sentiment, Behavior).
7. Confident Decision: User makes a safe, money-saving purchasing decision based on authentic feedback.

Rate yourself

3

Design

- 1 point – The use of AI is a good fit for the solution.
- 2 points - The use of AI is a good fit for the solution and there is some documentation about how it meets the needs of users
- 3 points - The use of AI is a good fit for the solution. The new user experience is clearly documented

7. Data

7.1 What data will you need to train your AI solution?

We need 40,000+ labeled customer reviews categorized into:

? Genuine Reviews (OR): Authentic customer experiences with balanced pros, cons, and usage details.

? Fake Reviews (CG): Computer-generated or paid reviews with repetitive praise and lack of specifics.

? Metadata: Star ratings (1-5?) and product categories (Electronics, Home, Books, Sports, Clothing).

7.2 Where or how will you source your data?

Data needed	Where will the data come from?	Who owns the data?	Do you have permission to use the data?	Ethical considerations
Have	Kaggle Fake Reviews Dataset (40,432 labeled reviews)	Kaggle & Academic Researchers	Yes ? Open Data Commons / CC license for research	Fully anonymized; zero PII; balanced 50/50 to eliminate bias.
Want/Need	Live storefront reviews from Amazon & Flipkart	Public e-commerce storefronts	Yes ? Publicly visible for fair-use academic analysis	Processed in temporary RAM; zero user profiling; rate-limited.
Nice to have	Regional reviews in Hindi & Hinglish	School community respondents	Yes ? User consent via survey forms	Parental consent for minors; strict survey confidentiality.

Rate yourself

3

Data

1 point – Relevant data to train the AI model have been identified as well as how the data will be sourced or collected.

2 points - Relevant data to train the AI model have been identified as well as how the data will be sourced or collected. There is evidence that the dataset is balanced.

3 points - Relevant data to train the AI model have been identified as well as how the data will be sourced or collected. There is evidence that the dataset is balanced, and that safety and privacy have been considered.

8. Prototype

8.1 Which AI tool(s) will you use to build your prototype?

Python 3.11, Scikit-Learn (TfidfVectorizer, LogisticRegression, MultinomialNB), Pandas, NumPy, and Pickle.

Flask REST API (api/index.py, app.py), BeautifulSoup4 + Selenium (scraper.py), Next.js, and Tailwind CSS on Vercel.

8.2 Which AI tool(s) will you use to build your solution?

8.3 What decisions or outputs will your tool generate and what further action needs to be taken after a decision is made?

1. Binary Review Classification: Flags each review as 'Genuine' or 'Fake' with calibrated confidence %.
2. Adjusted Rating Recalculation: Filters deceptive reviews to expose the true genuine product star rating.
3. Trust Score Output: 80-100 -> Safe to buy; 50-79 -> Caution advised; <50 -> High deception risk, avoid purchase.

Rate yourself

3

Prototype

1 point – A concept for a prototype shows how the AI model will work.

2 points - A prototype for the solution has been created and trained.

3 points - A prototype for the solution has been created and successfully trained to meet users' requirements

9. Testing

9.1 Who are the users who tested the prototype?

12 test users: 6 high school students (Class 11 & 12), 4 parents who shop online, and 2 school faculty members (Computer Science & Science).

Users were shocked when popular 4.5? products dropped below 3.5? after filtering fake reviews. Non-technical users praised the visual Trust Score gauge and single-review forensics breakdown.

9.2 List your observations of your users as they tested your solution.

9.3 Complete the user feedback grid

What works ➤ ? Instant sub-50ms review analysis. ? Clear Advertised vs Adjusted rating comparison. ? Intuitive 3-color palette (Green/Yellow/Coral).	What needs to change ➤ . ? Simplify technical ML jargon for consumers. ? Eliminate duplicate navigation buttons. ? Clarify CAPTCHA fallback instructions.
Questions? ? 'Does the AI detect sarcasm in reviews?' ? 'Can I check reviews on mobile while shopping?' ? 'Will it support Hindi/Hinglish reviews?'	Ideas ➤ ? Build a 1-click Chrome browser extension. ? Add review burst velocity detection over time. ? Expand dataset to Indian regional languages.

9.4 Refining the prototype: Based on user testing, what needs to be acted on now so that the prototype can be used?

1. Streamlined Navigation: Removed duplicate dashboard buttons and simplified technical headings.
2. Robust CAPTCHA Fallback: Added clear prompt to paste text or upload CSV when Amazon blocks requests.
3. Enhanced Dual-Rating Visuals: Made the genuine adjusted star rating prominent alongside advertised stars.
4. Single-Review Tester: Added live reactive textarea with real-time risk factor tagging.

9.5 What improvements can be made later?

1. Chrome & Edge Browser Extension: Inject Trust Scores directly into Amazon & Flipkart shopping pages.
2. Review Burst Detection: Temporal analysis to flag unnatural review volume spikes posted on a single date.
3. Transformer Models: Fine-tune lightweight DistilBERT for nuanced contextual deception and sarcasm.
4. Multilingual Expansion: Train models on Hindi and Hinglish e-commerce reviews.

Rate yourself

3

Testing

- 1 point – A concept for a prototype shows how it will be tested.
2 points - A prototype has been tested with users and improvements have been identified to meet user requirements.
3 points - A prototype has been tested with a fair representation of users and all tasks in this section have been completed.

10. Team collaboration

10.1 How did you actively work with others in your team and with stakeholders?

Our team practiced agile collaboration led by Deshna (Project Leader & Coder) with daily 15-minute school check-ins and weekly sprint reviews on Google Meet. Bhavya (Researcher) and Charvi (Marketing Leader) conducted user surveys and translated pain points into project requirements. Aryan (Designer & Tester) created wireframes and user testing protocols. Darshit (Video Producer) documented all milestones on video. We consulted our computer science teacher, Ms. Priyanka Sharma, regularly for feedback on algorithmic fairness and model evaluation.

Rate yourself

3

Team collaboration

- 1 point – There is some evidence of team interactions among peers and stakeholders.
- 2 points - Team collaboration among peers and stakeholders is clearly documented in this section.
- 3 points - Effective team collaboration and communication among peers and stakeholders is clearly documented in this section..

11. Individual learning reflection

11.1. Team Reflections

A good way to identify what you have learned is to ask yourself what surprised you during the project. List the things that surprised you and any other thoughts you might have on issues in your local community.

Team Member Name: [Aneeshna \(Group Leader, Data Expert, Prototype Builder\)](#)

What surprised me most was how mathematical models can detect language deception. Achieving 89.4% accuracy with Logistic Regression showed me the power of NLP. Leading this project taught me how to manage sprint deadlines, balance datasets ethically, and integrate full-stack AI systems.

Team Member Name: [Aryan \(Designer & Tester\)](#)

I was surprised by how crucial UX design is for AI. An 89.4% accurate model is useless if users can't understand it. Designing the dual star rating comparison and Trust Score gauge taught me how to transform complex machine learning probabilities into intuitive consumer protection tools.

Team Member Name: [Aneevya \(Information Researcher\)](#)

I was shocked by the scale of the fake review problem during user interviews? nearly every student and parent had been misled by a 4.5+ star product. Researching review spam algorithms proved that AI can serve as a vital consumer defense mechanism.

Team Member Name: [Anishil \(Video Producer\)](#)

I learned that technical engineering requires compelling storytelling. Translating complex TF-IDF vectorization and Python Flask APIs into an engaging 3-minute video pitch taught me how to communicate scientific innovations clearly to diverse audiences.

Team Member Name: [Anshu \(Marketing / Communication Leader\)](#)

I was amazed by our school community's enthusiastic response to TrustCheck. Managing our logbook documentation, user interviews, and pitch narrative taught me how to align technical development with real human needs and market viability.

Team Member Name:

Rate yourself

3

Individual Learning Reflection

- 1 point – Some team members present an account of their learning during the project.
- 2 points - Each team presents an account of their learning during the project.
- 3 points -Each team member presents a reflective and insightful account of their learning during the project

12. Video link

Enter the URL of your team video:

Code Link:

<https://github.com/muskanpapdiwal/TrustCheck>

Video Link:

<https://trust-check-rouge.vercel.app/> (Live App) | <https://youtu.be/TrustCheck-Demo>